Week 11 — Achievers / HOTS Mixed Practice — Answer Key

Q1. A) 12 — If the number is x: 3x = x + 24 → 2x = 24 → x = 12.

Q2. C) 25 — Three consecutive numbers summing to 72: let them be n−1, n, n+1; their sum is 3n = 72, so n = 24, and the largest is 25.

Q3. A) ₹1,750 — Perimeter = 2 × (20 + 15) = 70 m; cost = 70 × ₹25 = ₹1,750.

Q4. B) ₹600 — After spending 2/5, 3/5 of the money remains, and that equals ₹360. So the total = 360 × 5 ÷ 3 = ₹600.

Q5. C) 8 — 10 L = 10,000 mL; 10,000 ÷ 1,250 = 8 jugs.

Q6. C) 13 — Let the younger sister's age be y; the elder is y + 4. y + (y+4) = 22 → 2y = 18 → y = 9, so the elder is 13.

Q7. B) 250 kg — 3/10 of 500 = 150 kg sold Monday; 1/5 of 500 = 100 kg sold Tuesday; total sold = 250 kg; remaining = 500 − 250 = 250 kg.

Q8. C) 16 — The gaps between terms increase by 1 each time (+1, +2, +3, +4, +5): 1, 2, 4, 7, 11, 16.

Q9. B) 88 — Number = (divisor × quotient) + remainder = (5 × 17) + 3 = 85 + 3 = 88.

Q10. C) 100 — Roses = 1/4 × 240 = 60; marigolds = 1/3 × 240 = 80; lilies = 240 − 60 − 80 = 100.

Q11. C) ₹300 — Amounts: 180, 220, 260, 300 (each ₹40 more than the last); check: 180+220+260+300 = 960. ✓

Q12. B) 10 cm — Perimeter = 2 × (length + breadth) = 48 → length + breadth = 24 → breadth = 24 − 14 = 10 cm.

Q13. C) 20 — Total = people ahead + people behind + Rani herself = 8 + 11 + 1 = 20.

Q14. B) 1,290 L — Combined rate = 250 + 180 = 430 L/hour; in 3 hours = 430 × 3 = 1,290 L.

Q15. B) 38 — Let the number be 10a + b, with a + b = 11. Reversing gives 10b + a, and (10b+a) − (10a+b) = 9(b−a) = 45 → b − a = 5. Solving with a + b = 11: a = 3, b = 8, so the number is 38. Check: reversed = 83, and 83 − 38 = 45. ✓

Scoring: 12+/15 is a strong Achievers-section result — that's genuinely olympiad-level performance at this stage. Below that, go back to whichever Phase-1 week the missed questions came from before attempting Sample Paper 1.